



SECOND SEMESTER 2025-26
COURSE HANDOUT

Date: 02.01.2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : ECE F243 / EEE F243 / INSTR F243
Course Title : SIGNALS AND SYSTEMS
Instructor-in-Charge : Prof. Pawan K. Ajmera
Instructor(s) : Prof. Pawan K. Ajmera and Prof. Ramesh Sambangi
Tutorial/Practical Instructors: Prof. Pawan K. Ajmera, Prof. S. M. Zafaruddin
Prof. Pankaj Arora, Prof. Dinesh Rano, Prof. Ramesh Sambangi

1. Course Description: This course is intended to provide comprehensive coverage of Signals and Systems, a fundamental subject of Electrical Engineering. The topics covered are Continuous-time and discrete-time signals and systems, convolution, properties of linear time-invariant (LTI) systems, Fourier series, Fourier transform, Laplace transform, z-transform, system analysis, frequency response, analog filters, sampling and reconstruction.

2. Scope and Objective of the Course: This course covers the fundamentals of signals and systems. It focuses on the essential transformations for signal-processing applications. In addition to the regular lecture series, the weekly tutorial session will help students better understand the concepts. Evaluation components include quizzes (conducted during tutorial hours) and exams. Students must review the following mathematical topics: calculus, vector analysis, coordinate systems, arithmetic and geometric progressions, probability, and complex variables.

3. Course Learning Outcomes:

- Analyze and classify signals (continuous-time and discrete-time) based on properties such as periodicity, symmetry, energy/power, causality, and dimensionality, and represent signals mathematically and graphically.
- Model and classify systems as linear or nonlinear, time-invariant or time-varying, causal or non-causal, stable or unstable, and verify system properties using analytical methods.
- Analyze Linear Time-Invariant (LTI) systems using impulse response, convolution (continuous and discrete), and difference/differential equation models to determine system output for given inputs.
- Students will be able to analyze, model, and simulate signals and systems in time and frequency domains using mathematical and computational tools, and apply these concepts to real-world engineering applications in communication systems, control systems, signal processing, biomedical engineering, and multimedia systems.

4. Textbooks

T1: B. P. Lathi, "Signal Processing & Linear Systems," Oxford University Press, 2nd edition, 2009.

T2: I. J. Nagrath and S. N. Sharan, "Signals and Systems," TMH, 2nd edition, 2009.

5. Reference Books:

R1: A. V. Oppenheim, A. S. Willsky with S. H. Nawab, Signals and Systems, Pearson India Education Services, Second Edition, 2015.

R2: M. J. Roberts, Signals and Systems: Analysis Using Transform Methods and MATLAB, Tata McGraw-Hill Publishing Company Limited, Second Edition, 2003.



6. Course Plan:

Module No.	Lecture #	Lecture Session	Reference	Learning outcomes
1	1	Introduction		
2	2-5	Classifications; Mathematical Representation; Elementary signals: Unit Impulse, Unit Step, Unit Ramp, and Exponential; Transformations of the Independent Variable.	T1-1.1 to 1.5 T1- 8.1 to 8.4 R1 -1.1 to 1.4	Understand the fundamental signals in continuous-time (CT) and discrete-time (DT) representations.
3	6-8	Basic System Properties (Causality, Stability, Time-Invariance, Linearity, Invertibility, systems with and without memory); Interconnections of systems.	T1-1.6 to 1.8 T1- 8.5 R1-1.5 to 1.6	Understand the various properties associated with CT and DT systems.
4	9-13	Unit Impulse Response; Convolution Sum and Convolution Integral Representation; Properties of LTI systems.	T1- ch2, ch9 R1- 2.1 to 2.4	Understand the concepts of convolution in CT and DT domains.
5	14-17	Fourier Series Representation for periodic signals and its convergence, Properties of Fourier Series.	T1- ch3, ch10 R1- 3.1 to 3.8	Understand the concepts of Fourier series for periodic signals in CT and DT domains.
6	18-21	CT Fourier Transform and its Properties; continuous-time LTI System characterizations using the differential equations.	T1- ch4, R1- ch4	Understand the concepts of Fourier Transform for CT signals.
7	22-25	Laplace Transform and its properties, Region of Convergence; LTI System characterizations.	T1- ch6 R1- ch9	Understand the concepts of stability concerning ROC for CT LTI systems.
8	26-29	Conversion from CT signals to DT signals using the sampling theorem; Aliasing, Reconstruction of Signals; Effect of Under Sampling.	T1- ch5 R1- 7.1 to 7.3	Understand the concept of Sampling.
9	30-33	DT Fourier Transform and its Properties; discrete-time LTI System characterization using the difference equations.	T1- ch10 R1- ch5	Understand the concepts of Fourier Transform for DT signals.
10	34-37	z-transform and its properties, Region of Convergence, LTI System characterization; Hardware implementations (Direct form-I and II)	T1-ch11 R1 - ch10	Understand the concepts of stability with respect to ROC for DT LTI systems.
11	38-40	Introduction to analog filters	R1-Ch6 Notes	Application of signals and systems

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30% (60 Marks)	09/03/2026 FN1	Open Book
Comprehensive Examination	180 Min.	45 % (90 Marks)	02/05/2026 FN	Closed Book
*Announced Quizzes (Best 6 of 8)	10 Min.	15% (30 Marks)	During Tutorial Hour	Closed Book
**Assignments		10 % (20 Marks)	TBA	



8. Chamber Consultation Hour: Announced in Class

9. Class Participation: Students are expected to attend all the classes and engage actively in class discussions. Attending class means engaging with ideas, collaborating with peers, and gaining essential insights, transforming learning into a dynamic experience that unlocks your potential. Each missed lecture is a missed opportunity to grow and ask questions.

10. Notices: Notices regarding the course will be displayed only on NALANDA.

11. Make-up Policy: Make-up is a one-time provision that applies only to Mid-semester and Comprehensive exams. Makeup will be granted only in cases of extreme emergencies, subject to the final decision of the instructor-in-Charge (IC)

12. Minimum performance requirement for a valid grade: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

***Announced Quizzes (Best 6 of 8):** Total 30 marks, 5 marks per quiz

- The 10-minute quiz will be held during the regular tutorial hours (Friday, 4:00 pm to 4:50 pm)
- The quiz date and topic will be announced on NALANDA.

****Assignments:** Total 20 Marks, 5 marks per assignment.

- In total, four assignments will be conducted, evenly spaced throughout the semester.
- Assignments will be conducted on the Nalanda portal.
- The assignment must be submitted by the deadline once it is announced.
- Assignments include a MATLAB task, which forms the basis for questions that need to be answered.

Material Allowed for OPEN BOOK exam:

- Self-Handwritten notes only.
- T1: B. P. Lathi, "Signal Processing & Linear Systems," Oxford University Press, 2nd edition, 2009.
- T2: I. J. Nagrath and S. N. Sharan, "Signals and Systems," TMH, 2nd edition, 2009.
- R1: A. V. Oppenheim, A. S. Willsky with S. H. Nawab, Signals and Systems, Pearson India Education Services, Second Edition, 2015.

Material NOT Allowed for OPEN BOOK exam:

- Photocopies

Please note that any material not specified above will be considered an act of academic malpractice.

Instructor-in-charge
Course No. ECE/EEE/INSTR F243